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IN PROGRESS | EE 31 JUNIOR DESIGN | FALL 2026

Autonomous ESP32 Robot

ESP32 | Arduino | IR LED & photodetector | PCB design | Onshape

PROBLEM

- The robot has to follow a colored lane and avoid obstacles without human input.
- It also has to receive commands from a partner team's robot.
- Every decision it makes depends on sensor readings it can trust.

SO FAR

- Programmed a 7-state state machine on the ESP32, advanced by a pushbutton interrupt or a websocket command.
- Selected the IR LED and photodetector resistor values and subtract the room's background light from each reading.
- Developing detection logic that filters noise in the color readings before the robot decides which lane color it is on.

NEXT

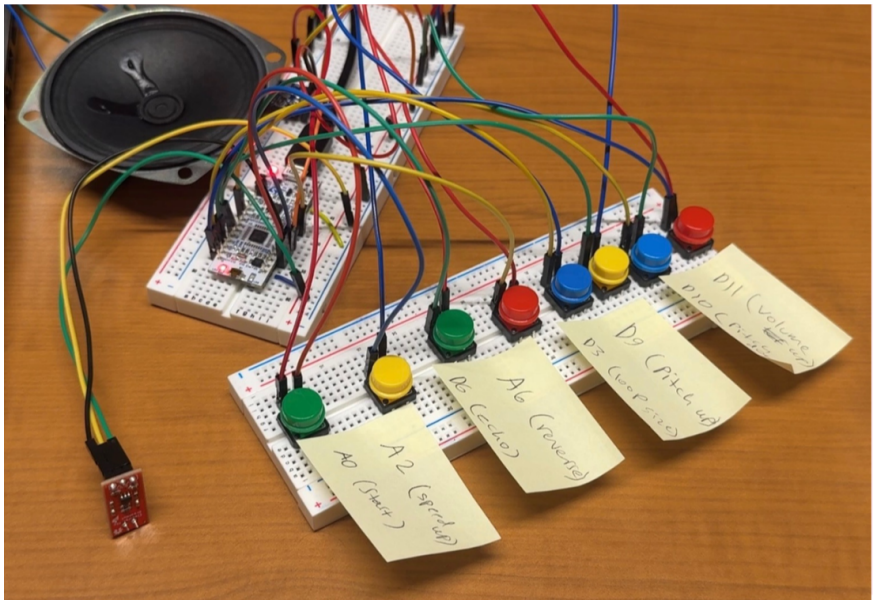
- Designing the robot's custom ESP32 PCB.
- Designing its outer body in Onshape.

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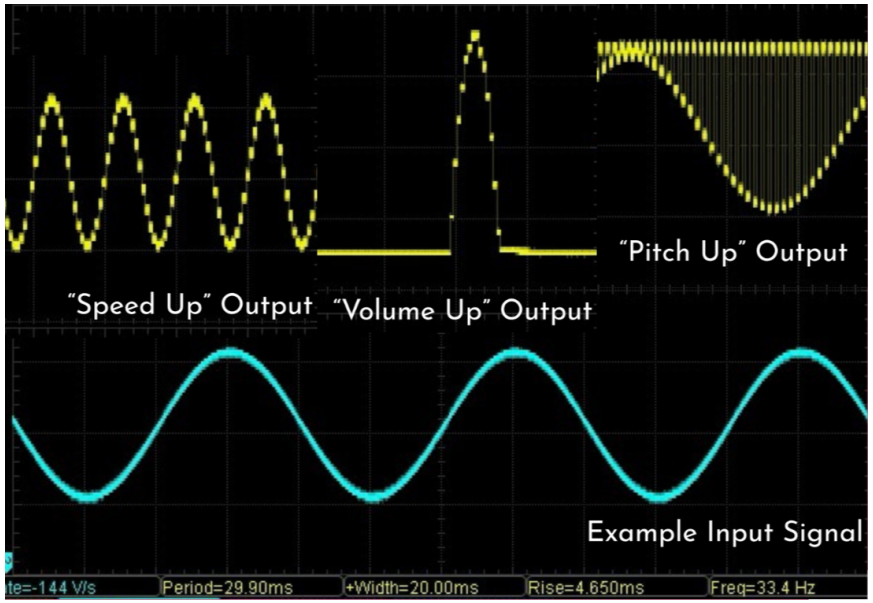
EE 14 EMBEDDED SYSTEMS | JAN. 2026 TO MAY 2026

DJ Board on an STM32

STM32L432KC | C | ADC & DAC | Oscilloscope



The board: microphone, speaker and effect buttons



Oscilloscope captures of the effect outputs

PROBLEM

- A real-time audio effects board records sound and remixes it with 7 effect buttons.
- Every effect depends on a clean recording.
- The first playback came out with a high-pitched hiss and clicks.

SOLUTION

- Wrote a timer-driven state machine so every sample is recorded and played at a steady rate.
- Gave each microphone reading more time to finish so no sample was cut short.
- Held the speaker at a steady resting level between clips and faded each clip in and out.

RESULT

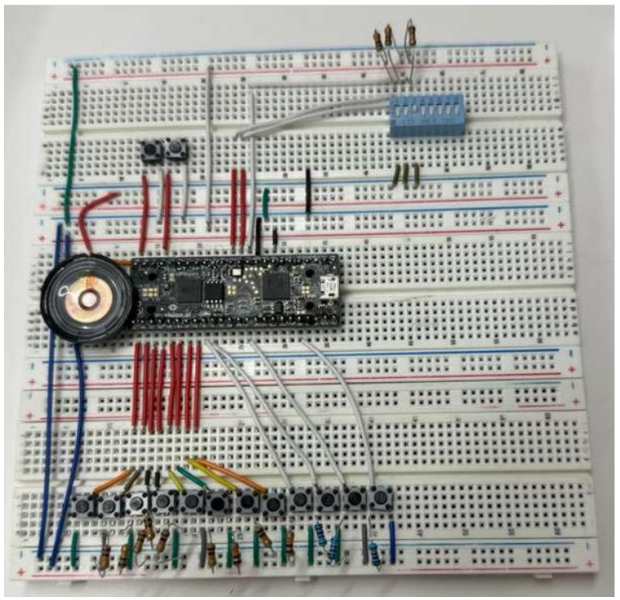
- An oscilloscope confirmed that the playback matched the original recording.
- All 7 effects run using only 64KB of on-chip memory.

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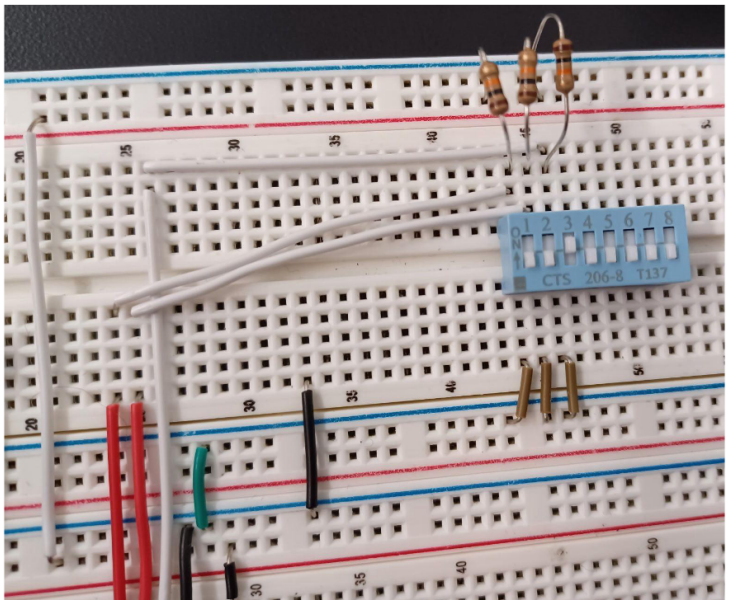
ES 4 DIGITAL LOGIC CIRCUITS | SEPT. 2025 TO DEC. 2025

FPGA Audio Synthesizer

SystemVerilog | FPGA | PWM



The full board



The switch that selects the octave

PROBLEM

- Each octave doubles a note's frequency.
- The first design calculated every note by dividing on the chip.
- Division needed over 250% of the FPGA's space.

SOLUTION

- Hand-calculated one octave and used bit shifts for the rest, since dividing by 2 is the same as shifting bits.
- Mixed the notes with PWM after testing showed that chords distorted.

RESULT

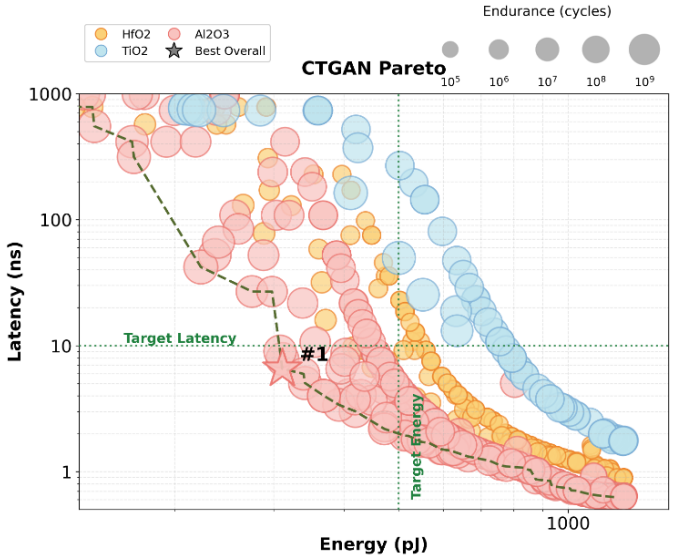
- An 8-octave synthesizer, from 16Hz to 8kHz, for a MIDI keyboard.
- 6 notes play clearly at once.

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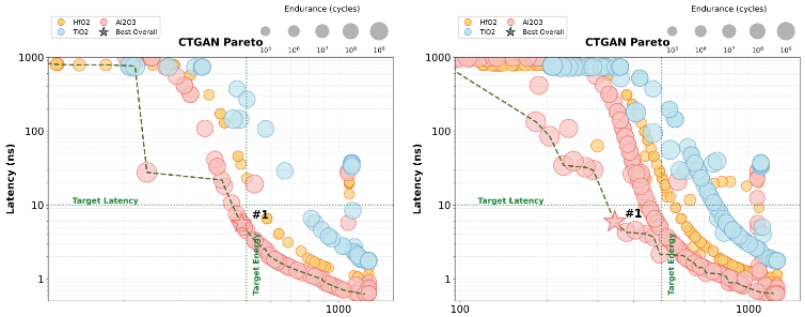
TUFTS EMERGING CIRCUITS AND SYSTEMS LAB | SEPT. 2025 TO APRIL 2026

CTGAN Design Discovery for RRAM

Python | PyTorch | pandas | CUDA | CTGAN | PINN



Final results with 5,000 training samples



The same search with 500 samples (left) and 2,500 (right)

PROBLEM

- A published AI framework designs RRAM memory devices.
- Its model averaged the training data, which smoothed away extreme designs.

SOLUTION

- Replaced the model with a CTGAN, which searches those extremes more aggressively.
- Scaled the training data 10 times, to 5,000 samples, so the CTGAN could learn reliably.
- Used the framework's physics model as a referee to reject impossible designs.

RESULT

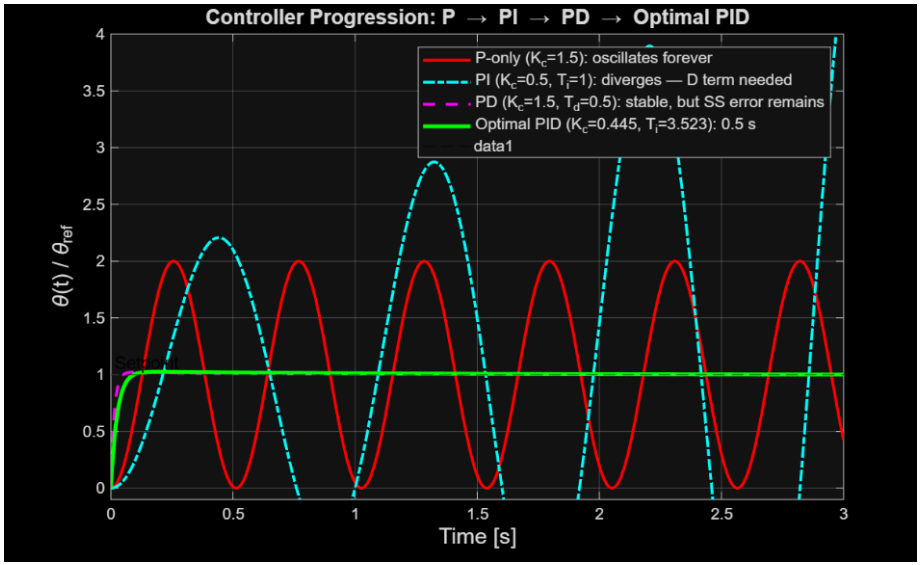
- 25% higher endurance than the published result.
- 11% lower switching energy.
- Slightly slower switching: 6.57ns vs 4.07ns.

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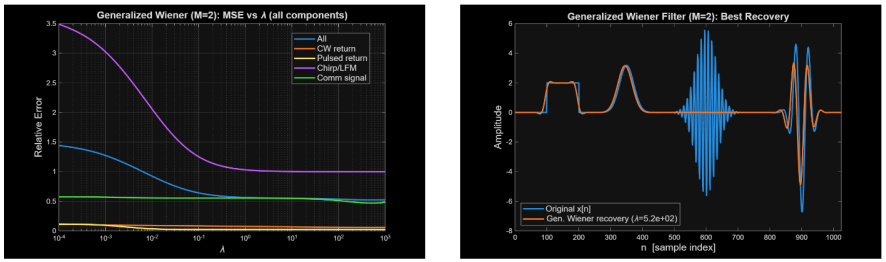
EE 23 LINEAR SYSTEMS | JAN. 2026 TO MAY 2026

UAV Pitch Control and RF Signal Restoration

MATLAB | PID control | Wiener filtering



P, PI, PD and PID controllers on the same UAV



Wiener filter error (left) and the restored RF signal (right)

PROBLEM

- Keeping a UAV level in pitch is the first step toward collecting clear RF signals from the air.
- Even a stable UAV shakes slightly and blurs the signals it receives.

SOLUTION

- Showed with Routh-Hurwitz analysis that P and PI controllers can never stabilize the UAV.
- Searched 15,625 gain combinations for the best PID settings.
- Restored a blurred, noisy RF signal with a Wiener filter.

RESULT

- 52ms rise time, 2.6% overshoot and 500ms settling time, within real UAV specs.
- 77% lower error than a direct inverse (MSE 2.272 to 0.521).